

How Stable is SHAP *under Oversampling*? A Study of Explanations in Imbalanced Scenarios

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Abstract Class imbalance is common in classification problems, yet it remains unclear to what extent traditional oversampling techniques alter the learned decision logic of classifiers and, consequently, the explanations derived from them. Given that SHapley Additive exPlanations (SHAP) is one of the most widely used post-hoc explanation methods, this paper examines how popular oversampling strategies affect the stability of SHAP explanations, focusing on robustness to training perturbations and sensitivity to oversampling intensity. We conduct a large-scale study on 25 imbalanced benchmark datasets, three tree-based classifiers (Random Forest, Gradient Boosting, and Extremely Randomized Trees), and four oversampling techniques (random oversampling, SMOTE, ADASYN, SVMSMOTE), at four resampling levels (0%, 15%, 30%, 50%) in the minority class, resulting in 48,750 fitted models. We find that increasing oversampling is associated with improved local SHAP stability in 85% of cases. However, this effect varies significantly across datasets and classification models, and no specific method consistently dominates. A meta-feature analysis further shows that instability is not explained by imbalance ratio and is only weakly related to predictive performance; instead, it is most associated with dataset structure, increasing with stronger low-order co-occurrence patterns and sparsity, and decreasing with higher attribute entropy.

Keywords: Imbalanced · XAI · Explainable Artificial Intelligence · Interpretability · SMOTE · Tree-based Ensemble Model · SHAP

1 Introduction

Machine learning is now routinely applied in a vast domains, where errors may have substantial social or economic impact and accuracy alone is not an adequate guarantee of reliability [5, 16]. In these settings, eXplainable Artificial Intelligence (XAI) has emerged as a way to render complex models accessible to diverse stakeholders, including engineers, domain specialists, and regulators [5, 25]. Recent surveys have introduced taxonomies of XAI techniques and associated quality criteria, arguing that useful explanations should be faithful to the underlying model, consistent across similar situations, supportive of human scrutiny and diagnosis, and coherent with the properties of the training data [32, 42]. Such requirements become critical when explanations are employed to inform high-stakes decisions or to inspect learned models for potential biases.

Among post-hoc model agnostic methods, SHapley Additive exPlanations (SHAP) has become one of the most widely adopted techniques for explaining individual predictions [28]. SHAP provides local additive produces local, additive feature attributions grounded in Shapley values, and these attributions satisfy several desirable axioms: local accuracy (the attributions sum to the model output relative to a baseline), missingness (features that are absent receive zero attribution), and consistency (if a feature’s marginal contribution to the prediction increases in a model change, its attribution does not decrease) [27, 28]. Global views can then be obtained by aggregating these local explanations across instances, for example by averaging absolute attributions or summarising their distribution over the dataset [28]. As a consequence, SHAP is often used as a default post-hoc explainer and is often treated as providing robust, model-faithful attributions even under class imbalance. Several studies indicate that its behaviour can, in fact, be sensitive to the class distribution and to the use of rebalancing techniques [7, 11, 34, 38].

Shapley-based importance profiles and local attributions have been shown to shift when the minority class is strengthened, or synthetic samples are introduced, with features gaining or losing apparent relevance as oversampling or other countermeasures are applied [7, 11]. This raises the possibility that, in imbalanced scenarios, SHAP explanations may not be as stable or invariant to common resampling strategies as often assumed, and motivates a systematic investigation of their stability and consistency when class imbalance is explicitly addressed.

Specifically, at the model level, stability can be quantified as the variability of outputs across independent runs with the same data and hyperparameters. When this notion of stability is considered in practical pipelines, a key source of variation is how we handle skewed data distributions. Class imbalance is a pervasive characteristic of real-world datasets and has a well-documented impact on learning algorithms and evaluation metrics [21], so any change in the way imbalance is addressed may also affect the stability of both predictions and their associated explanations.

This kind of change is necessary when a minority class is severely under-represented [19]. Standard classifiers tend to favour the majority class, which

motivates the use of resampling strategies such as SMOTE and its variants to rebalance the data [8, 21]. Oversampling methods interpolate or synthesise new minority instances in feature space and therefore alter the geometry of the training set. From a predictive perspective, this can improve minority class performance. From an XAI perspective, it raises an important question that directly touches the notion of dataset consistency: to what extent do post-hoc explanations, such as SHAP, remain stable when the training data are artificially modified to correct class imbalance?

Recent XAI surveys emphasise that explanations should be analysed not only in terms of their capability to approximate model behaviour, but also with respect to how robust they are to modelling choices, data manipulations, and hyperparameter changes [22, 32]. Vilone and Longo explicitly discuss consistency and faithfulness as dimensions along which explanation methods should be evaluated, and identify dataset-related consistency as a separate requirement that captures how explanations react to changes in the data-generating or data-processing pipeline [42].

To address this gap, we study SHAP under imbalanced classification pipelines that explicitly correct class skew via standard oversampling, using a large-scale design to account for the strong dataset- and seed-dependent variability typical of imbalanced learning. Our experimental design spans 25 benchmark datasets [10], three tree-based classifiers: Random Forest (RF), Gradient Boosting (GB), and Extremely Randomized Trees (ERT), and four oversampling methods: Random Oversampling (ROS), Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), and Support Vector Machine SMOTE (SVMSMOTE). For all combinations, we consider four resampling levels (0%, 15%, 30%, and 50%). For each dataset, classifier and oversampling algorithm configuration, we train five seed variants and compute SHAP vectors on a fixed subset of test instances.

Also, rather than assessing SHAP’s axiomatic *consistency*, which is defined with respect to changes in a model’s underlying set function, we focus on *stability* as an operational property of the full learning pipeline, capturing how explanations vary under practical perturbations such as random initialisation and oversampling. Accordingly, we introduce an operational stability indicator by computing the mean cosine distance between SHAP attribution vectors across seeds and summarising, for each configuration, how this distance trends as oversampling intensity increases. Finally, to clarify when explanation drift is more likely to occur, we relate the observed instability to intrinsic dataset characteristics by extracting a broad set of meta-features and analysing their association with instability patterns across datasets.

Our analysis is guided by three research questions:

- RQ1: How do local SHAP explanations change as class balancing in different increased by oversampling?
- RQ2: Is it possible to identify a classifier and resampling combination that yields more stable explanations across different random initialisations?

- RQ3: Which intrinsic dataset characteristics are associated with SHAP instability under oversampling?

By answering these questions, we characterise SHAP stability under oversampling in imbalanced settings and relate it to XAI requirements such as consistency, and dataset characteristics.

The remainder of the paper is organised as follows: Section 2 reviews related work on explanation stability and imbalanced learning, Section 3 describes the experimental design and instability metric, Section 4 reports the empirical results and meta-feature analysis, Section 6 discusses limitations and extensions, and Section 7 concludes the paper.

2 Related Work

Work at the intersection of class imbalance, SMOTE-like oversampling and post-hoc explainability can be broadly divided into two strands: studies that document how balancing methods alter model behavior and explanations, and studies that adopt SHAP as a reliable explainer on imbalanced data and implicitly assume robustness.

From the imbalance side, a large body of work has characterized how skewed class distributions affect learning algorithms, evaluation metrics and data complexity [13, 20, 46]. Data level resampling (balancing methods) such as SMOTE [8], ADASYN [18] and their numerous variants are now standard tools for oversampling the minority class [17]. These methods are known to modify the geometry of the feature space, potentially introducing synthetic points that do not faithfully reflect the original data distribution and may shift decision boundaries [17]. Recent surveys emphasize that, beyond predictive gains, there is still a need to understand why, when and how class imbalance countermeasures work and how they interact with the goals of interpretability and knowledge extraction [7, 9].

A first group of works explicitly reports that balancing methods can change explanations and feature importance profiles. Cemernek et al. systematically evaluate class imbalance countermeasures and show that both resampling and algorithmic strategies can jeopardize interpretability when models are used to derive knowledge, with rebalanced models exhibiting shifted concepts and altered variable importance patterns [7]. Menzies et al. and subsequent studies on defect prediction find that class rebalancing techniques modify the learned concepts, as evidenced by changes in feature importance and partial dependence plots, even when performance improves [7]. Staño et al. analyzed the effect of several under and oversampling methods (including SMOTE, SMOTETomek and Borderline SMOTE) on partial dependence profiles and propose stability measures based on summary distribution distances, concluding that balancing can substantially alter model behavior as seen through XAI tools [40]. At the level of specific XAI methods, Patil et al. investigate the “explainability of SMOTE based oversampling” and observe that SMOTE variants change both performance and the

prominence of explanatory features in tabular problems [34]. Sahlaoui et al. combine SMOTE variants with LIME and SHAP for student performance prediction and report that different oversampling strategies lead to different local and global importance profiles, even when accuracy is comparable [38]. Similar concerns appear in domain specific studies: Nayan et al. use SMOTE and NearMiss for diabetes diagnosis with XAI visualizations and show that rebalancing affects which risk factors are highlighted [33], while work on adverse drug reaction detection and stroke prediction explicitly relies on resampling to improve minority recall but does not obtain invariant explanations across balancing configurations.

More specifically for SHAP and tree based models, Dube and Verster analyze random forests for churn and fraud detection under varying levels of class imbalance and report that Shapley based feature importance profiles change systematically as the minority proportion is increased [11]. Some predictors gain or lose apparent relevance when the class distribution is rebalanced, and individual level breakdown plots reveal substantial shifts in the contributions assigned to the same instance across class ratios [11]. Their results suggest that interpretations derived from Shapley values become less stable as the degree of imbalance grows, and that explanation reliability must be read in the context of class ratios. Alarab and Prakoonwit similarly show that data resampling strategies can alter feature importance rankings in blockchain fraud detection, even when the underlying classifier is unchanged [1]. Together with broader empirical evidence that different XAI techniques often disagree on the same model and task [37], these studies point to a non trivial interaction between class imbalance, rebalancing methods and post-hoc explanations.

A second strand of work treats SHAP as a practical default for explaining trained classifiers, including in settings where the training data are class-imbalanced. In applied pipelines, especially in healthcare, SHAP is commonly used to support model justification by reporting local attributions and aggregated feature rankings, even when the underlying learning problem is explicitly imbalanced and handled with rebalancing steps [4, 44]. Comparative studies of local interpretability techniques in medical settings further reinforce this positioning by reporting that SHAP explanations are often competitive in terms of fidelity and are generally well received by domain experts, making it a frequent baseline choice for tabular models [12]. At the methodological level, surveys and taxonomies highlight SHAP as a principled attribution approach grounded in cooperative game theory, with desirable properties such as local accuracy and consistency that contribute to its broad adoption [5, 28, 42]. Benchmarking work that evaluates post-hoc explainers against ground-truth explanation datasets also typically includes SHAP as a strong reference method [30]. However, much of this evidence is derived from balanced or mildly imbalanced benchmarks, and the sensitivity of SHAP explanations to explicit class rebalancing and oversampling choices is still not systematically characterized, which motivates the experimental analysis in this paper.

Beyond specific methods, several meta evaluation frameworks emphasize stability and consistency as key desiderata for trustworthy explanations. Vilone

and Longo identify properties such as fidelity to the underlying model, internal consistency, scrutability for human users and dataset related consistency as central for responsible XAI [42]. Quantitative proposals such as stability indices for LIME [43], the strength of explanation framework [39] and the MetaQuantus suite for meta evaluating explanation metrics [3] all highlight that explanations should not fluctuate arbitrarily under small perturbations of the learning pipeline. Dablain et al. argue that, in imbalanced learning specifically, there is still a scarcity of interpretability techniques adapted to skewed data and call for frameworks that jointly reason about data complexity, imbalance and explanations [9]. Yet, despite this emphasis on stability, there is little systematic evidence on how SHAP attributions evolve when standard oversamplers such as ROS, SMOTE, ADASYN and SVM-SMOTE are applied at different resampling levels.

In summary, existing work shows that class imbalance and its countermeasures can influence model interpretability and that SMOTE based oversampling is both widely used and potentially distortionary. At the same time, SHAP is often treated as a stable, high quality explainer even on imbalanced data, and most empirical studies either focus on performance or assess explanations in a single balancing configuration. To the best of our knowledge, there is no systematic, multi dataset study that quantifies the stability of SHAP explanations with respect to oversampling level and algorithm, under controlled repetitions of the training process. Our work addresses this gap through a controlled experimental methodology that evaluates SHAP across multiple benchmark datasets, tree-based classifiers, and oversampling strategies, while explicitly characterizing dataset-specific properties via meta-features to assess how intrinsic data characteristics influence explanation stability.

3 Methodology

3.1 Overall Experimental Design

The research questions addressed in this paper require systematic comparisons across datasets, oversampling methods, and classifiers to assess overall trends and to identify configurations that yield more stable explanations. Accordingly, we design a factorial study comprising 25 imbalanced binary classification datasets with heterogeneous properties (336–20,000 instances, 6–617 features, and imbalance ratios from 8.6:1 to 130:1), three classifiers (XTR, RF, and GBC), four oversampling methods (ADASYN, ROS, SMOTE, and SVMSMOTE), and four resampling levels (0%, 15%, 30%, and 50%). Figure 1 summarizes how these components are combined in the experimental design.

Stability involves an implicit comparison because it quantifies how SHAP values change due to model perturbations. To evaluate the changes between two different situations, it is necessary to establish a stability baseline. We first evaluated the stability for a group of classifiers that differ only by randomization, as done in [23]. We chose to fit 5 classifiers that differ only by randomization for

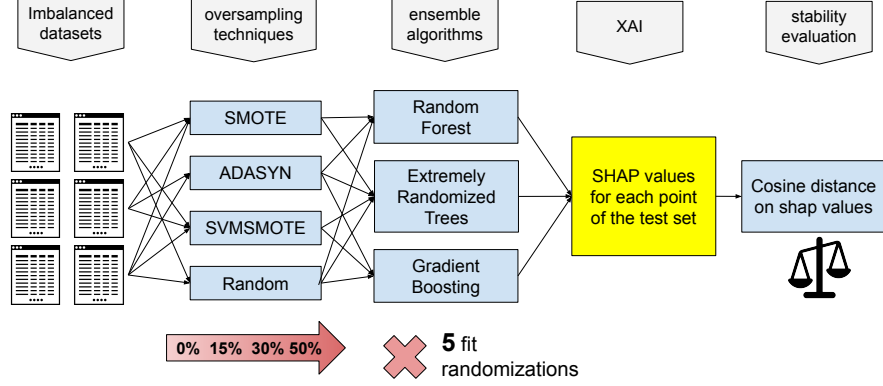


Figure 1: Overall experiment configuration.

each of the 975 experimental configurations, resulting in a total of 4,875 classifiers fitted for the experiment. The experiment was repeated 10 times changing randomizations at different levels for 48,750 total fitted models. Although we presented the data of only one experiment, all the other fully confirmed these results.

To quantify attribution stability, we adopt the cosine distance between local SHAP attribution vectors as an operational measure of explanation disagreement, following the SHAP Distance formulation proposed by Yu et al. [48], and compute it across repeated training runs (different seeds) for the same test instances. For a given dataset–classifier–oversampler scenario and oversampling level $\ell \in \{0, 15, 30, 50\}$, let $\phi_{i,\ell}^{(s)} \in \mathbb{R}^p$ denote the SHAP vector for test instance i produced by the model trained with seed $s \in \{1, \dots, S\}$ (with $S = 5$). We define the instance-level instability at level ℓ as the mean pairwise cosine distance across seeds,

$$d_{i,\ell} = \frac{2}{S(S-1)} \sum_{1 \leq s < t \leq S} \left(1 - \frac{\phi_{i,\ell}^{(s)} \cdot \phi_{i,\ell}^{(t)}}{\|\phi_{i,\ell}^{(s)}\| \|\phi_{i,\ell}^{(t)}\|} \right). \quad (1)$$

To characterise the effect of increasing oversampling, we compute a trend statistic over ℓ , such as a slope of $d_{i,\ell}$ across levels or the contrast $\Delta_i = d_{i,50} - d_{i,0}$. Finally, each scenario is summarised by the proportion of the $n = 50$ test instances with a positive trend (e.g., $\Delta_i > 0$), which we report as an *instability indicator*; by construction, larger values indicate worse stability.

In our setting, *instability* is defined as the complement of stability: larger cosine distances (and increasing trends with oversampling) indicate greater disagreement among explanations across repeated trainings, hence lower stability.

The entire process was applied across 300 different scenarios, combining 25 datasets with 3 oversampling techniques and 4 classifiers. The final result for the 300 different scenarios is the percentage of the 50 observations considered in

which SHAP instability increases with oversampling (evaluated as relative cosine distance increment).

In each scenario (dataset, classifier and oversample method), we perform a stratified train/test split, generate three oversampled training sets at 15%, 30%, and 50% (plus the 0% baseline), train five seed variants per level, and compute SHAP vectors on the same 50 test instances; stability is measured as the mean pairwise cosine distance across seeds at each level and summarised by the fraction of instances whose distance increases with oversampling (instability indicator).

3.2 Imbalanced Datasets

To evaluate the effectiveness and robustness of the considered oversampling methods, we conducted experiments on a diverse collection of imbalanced benchmark datasets. In particular, we selected 25 datasets from the study of [10] and the *imbalanced-learn* library [24], which are widely used in the imbalanced learning literature.

These datasets span a broad range of application domains, including biology, medicine, remote sensing, image recognition, and text processing, thereby ensuring that the experimental evaluation reflects realistic and heterogeneous learning scenarios. Moreover, they exhibit substantial variability in terms of class imbalance, sample size, and feature dimensionality.

The imbalance ratios range from moderately imbalanced datasets, such as *ecoli* (8.6:1), to extremely skewed datasets, such as *protein_homo* (111:1) and *abalone_19* (130:1). The number of instances varies from a few hundred samples (e.g., *arrhythmia* and *libras_move*) to 20,000 samples (*letter_img*), while the number of features ranges from low-dimensional representations (6 features in *mammography*) to high-dimensional datasets with hundreds of attributes (617 features in *isolet*).

This wide variability allows us to assess the behaviour of oversampling techniques under different levels of data sparsity, redundancy, and complexity. In particular, high-dimensional and highly imbalanced datasets pose significant challenges for both classification performance and model stability, making them well suited for studying the impact of data augmentation on predictive accuracy and explanation robustness.

Table 1 summarizes the main characteristics of the datasets, including their imbalance ratio, number of samples, and number of features. Together, these datasets provide a comprehensive benchmark for evaluating the proposed methodology across a wide spectrum of imbalanced learning scenarios.

Meta-features extraction Meta-features are quantitative descriptors that characterize the structural, statistical, and informational properties of datasets. They are widely used in meta-learning and algorithm selection to model relationships between dataset characteristics and learning performance [35, 41]. By summarizing relevant aspects of data distributions, meta-features enable the analysis of how intrinsic dataset properties influence predictive accuracy, model behavior, and learning stability.

Table 1: Summary of the imbalanced benchmark datasets used in the experiments, sorted by increasing imbalance ratio. The table reports the imbalance ratio, number of samples, and number of features for each dataset.

Name	Ratio	#Samples	#Features
ecoli	8.6:1	336	7
optical_digits	9.1:1	5,620	64
satimage	9.3:1	6,435	36
pen_digits	9.4:1	10,992	16
abalone	9.7:1	4,177	10
sick_euthyroid	9.8:1	3,163	42
spectrometer	10.8:1	531	93
car_eval_34	11.9:1	1,728	21
isolet	12.0:1	7,797	617
us_crime	12.3:1	1,994	100
yeast_ml8	12.6:1	2,417	103
scene	12.6:1	2,407	294
libras_move	14.0:1	360	90
thyroid_sick	15.3:1	3,772	52
coil_2000	15.8:1	9,822	85
arrhythmia	17:1	452	278
solar_flare_m0	19:1	1,389	32
oil	22:1	937	49
car_eval_4	26:1	1,728	21
wine_quality	26:1	4,898	11
letter_img	26:1	20,000	16
yeast_me2	28:1	1,484	8
ozone_level	34:1	2,536	72
mammography	42:1	11,183	6
abalone_19	130:1	4,177	10

In this work, meta-features are employed to investigate the relationship between dataset characteristics and the instability of feature attribution methods under different oversampling strategies. Specifically, we analyze how variations in data complexity, sparsity, and information content are associated with fluctuations in explanation reliability after data augmentation.

The selected meta-features follow the taxonomy proposed by [26], which categorizes features into general, information-theoretic, statistical, complexity, itemset-based, concept, and landmarker groups. This taxonomy has been widely adopted due to its interpretability and effectiveness in characterizing data distributions across diverse learning scenarios [36, 41].

In total, 179 meta-features were extracted using the *PyMFE* framework version 0.4.4 [2]. The resulting meta-feature representation enables a systematic correlation analysis between dataset properties and explanation instability, providing insights into how intrinsic data characteristics influence the robustness of post-hoc interpretability methods in imbalanced learning scenarios.

3.3 Resampling Methods

To address the class imbalance present in the selected datasets, we employed four widely used oversampling techniques: ROS, SMOTE, ADASYN, and SVMSMOTE. These methods represent different strategies for generating minority class samples and are commonly adopted as baselines in imbalanced learning research.

ROS balances the class distribution by randomly replicating existing minority class instances. Although simple to implement, this approach may increase the risk of overfitting, as no new information is introduced into the training data [29].

SMOTE generates synthetic minority class samples by interpolating between existing minority instances and their nearest neighbours in the feature space. By creating new samples rather than duplicating existing ones, SMOTE aims to reduce overfitting while improving class balance [8, 47].

ADASYN extends the SMOTE framework by adaptively generating more synthetic samples in regions where the minority class is sparsely represented or harder to learn. This allows the algorithm to focus on more complex decision boundaries, potentially improving classifier performance on difficult examples [31].

SVMSMOTE further refines the sample generation process by incorporating information from a Support Vector Machine (SVM). New minority samples are generated near the decision boundary estimated by the SVM, with the goal of improving class separability and enhancing classification performance [45].

We implemented oversampling using the `imbalanced-learn` library (v0.14.0) with the standard classes `RandomOverSampler`, `SMOTE`, `ADASYN`, and `SVMSMOTE`. All methods were used with their default parameter settings to avoid confounding effects from method-specific tuning. This selection spans random replication, interpolation-based synthesis, adaptive generation, and boundary-aware synthesis, enabling a systematic assessment of how different augmentation mechanisms interact with classification performance and explanation stability.

3.4 Classification Algorithms

To evaluate the impact of different oversampling techniques on predictive performance and explanation stability, we employed three widely used tree-based ensemble classifiers: RF, GB, and ETR. These models are known for their strong performance on structured tabular data and their robustness to noise and feature interactions, making them suitable for imbalanced classification tasks.

RF is an ensemble learning method that constructs multiple decision trees using bootstrap sampling and random feature selection at each split. The final prediction is obtained by aggregating the outputs of individual trees, typically through majority voting. This combination of bagging and feature randomness reduces variance and mitigates overfitting, leading to stable and reliable predictions [6].

GBC builds an ensemble of decision trees in a sequential manner, where each new tree is trained to correct the errors of the previously constructed ensemble. By optimizing a loss function through gradient descent in function space, Gradient Boosting is able to model complex non-linear relationships and often achieves high predictive accuracy, particularly in challenging learning scenarios [14].

XTR Extra Trees further increase ensemble diversity by introducing additional randomness during tree construction. In contrast to Random Forest, splits are selected randomly rather than by optimizing an impurity measure, and the entire training set is typically used without bootstrapping [15]. This enhanced randomness reduces variance and computational cost, while maintaining competitive predictive performance.

We trained all models using the `scikit-learn` library (v1.72), using standard implementations `RandomForestClassifier`, `GradientBoostingClassifier`, and `ExtraTreesClassifier`. All classifiers were used with their default parameter settings to avoid confounding effects from hyperparameter tuning. Together, these learners cover complementary ensemble paradigms (bagging, boosting, and extreme randomisation), enabling a systematic evaluation of how different decision-structure inductive biases interact with oversampling and affect both predictive performance and explanation stability in imbalanced settings.

3.5 SHAP Implementation

The Python `shap` library (v0.50.0) was used to compute Shapley values. Since all evaluated predictors are tree-based ensembles trained with default hyperparameters, we employed `shap.TreeExplainer`, which leverages model structure to compute exact or fast approximate Shapley values for trees, avoiding the higher computational cost of model-agnostic explainers while preserving the local accuracy property of SHAP [27].

4 Results

This section first introduces the proposed instability indicator and reports its distribution across datasets, classifiers, and oversampling methods; it then progressively addresses RQ1–RQ3 by analysing how instability evolves with increasing oversampling intensity, comparing classifiers–oversampler combinations, examining its relationship with predictive performance, and relating the observed patterns to intrinsic dataset characteristics via meta-features, before concluding with practical implications for auditing-oriented pipelines.

4.1 Experimental outcomes and instability indicator

We summarise explanation stability across oversampling (Section 3.1) using an *instability indicator* derived from cosine distances between SHAP attribution

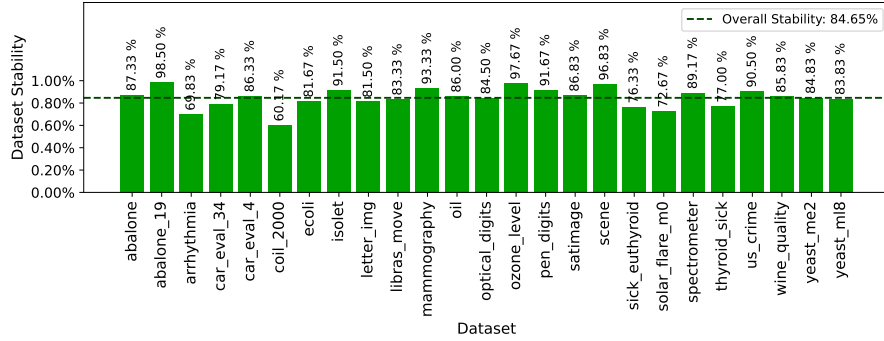


Figure 2: Dataset-level stability summary, reported as the percentage of test instances whose mean cosine distance between SHAP vectors does not increase as oversampling intensity rises (higher values indicate more stable behaviour).

vectors. Considering this, in our definition, *higher values indicate worse stability*: the indicator represents the proportion of test instances for which the mean cosine distance across seeds increases as the oversampling level grows. Thus, an indicator close to 0 means that almost no instances exhibit increasing cosine distance under stronger oversampling (stable behaviour), while larger values mean that a higher fraction of instances become less stable as resampling intensifies.

4.2 Effect of oversampling intensity on explanation stability

Figure 2 shows the dataset-level stability aggregated across all classifiers and oversampling algorithms. The distribution is heterogeneous, with some datasets exhibiting low instability and others showing substantially larger values, indicating that dataset-specific factors play a strong role in explanation variability under oversampling. The overall result is that in about 85% of considered observations, SHAP stability increases as the oversampling level moves from 0% to 50%, that SHAP stability improves across augmented levels of oversampling.

As discussed in the following sections, the average value of about 85% has high variability over diverse combinations of datasets, resampling, and classifiers. Several scenarios exhibited stability improvements in 100% of observations when oversampling was applied, whereas a few combinations showed SHAP stability decreases in 100

4.3 Stability differences across classifiers and oversamplers

To assess whether instability is primarily driven by the learning algorithm, Figure 3 aggregates the instability indicator by classifier. GBC exhibits the highest average instability, RF shows the lowest, and XTR lies between these two. This pattern is consistent with the fact that the learning algorithms expose different

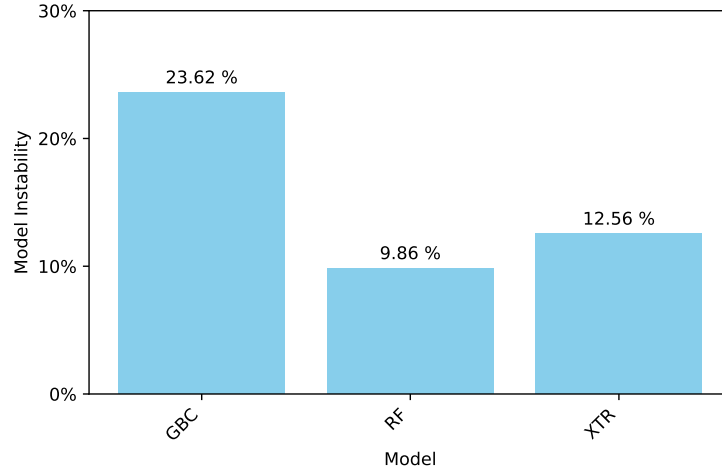


Figure 3: Model-level instability indicator (higher is worse).

sources of variability to the pipeline: boosting constructs an additive sequence of models in which early split decisions and residual updates can be amplified across iterations, whereas RF averages a large set of decorrelated trees, which tends to dampen the effect of small perturbations in the training data. XTR introduces stronger randomization at split selection than RF, but still benefits from ensemble averaging, which can explain its intermediate behaviour.

These differences have implications for potential bias under imbalance correction. Oversampling modifies the empirical distribution seen during training, particularly in regions occupied by the minority class. If a learner is more sensitive to such distributional changes, it may exhibit larger run-to-run variability in the learned decision structure near minority regions, and this can propagate to explanation variability for the same test instance. In this sense, higher instability may be interpreted as a warning signal for procedural sensitivity.

Considering the resampling methods, Figure 4 aggregates the instability indicator by oversampling strategy. Although differences are observed, they are smaller than the variation across datasets and classifiers, suggesting that explanation stability is driven more strongly by the learning problem and the model family than by the choice of oversampler alone. In this aggregate view, SVM SMOTE exhibits the lowest instability, whereas ROS and SMOTE tend to show higher values. However, because oversampling methods interact with the geometry of each dataset and with the inductive bias of each classifier, these averages should not be interpreted as a universal ranking.

In response to RQ2, we find that a consistently more stable choice can be identified at the *classifier* level but not as a universally optimal classifier and oversampler pair. RF yields the lowest average instability across random initialisations, with XTR presenting intermediate values and GBC exhibiting the high-

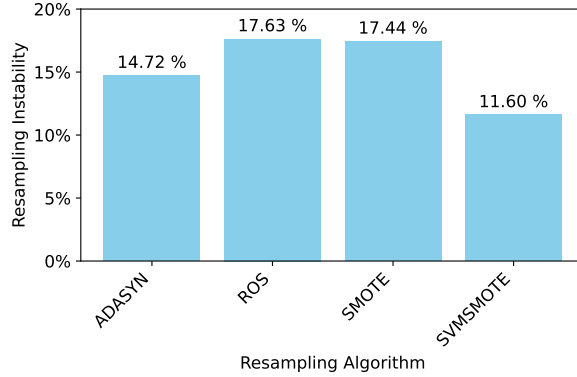


Figure 4: Oversampler-level instability indicator (higher is worse).

est instability (Figure 3). In contrast, differences among oversampling methods are smaller than the cross-dataset and cross-model variation (Figure 4); although SVMSMOTE achieves the lowest aggregate instability, oversample algorithms averages are not sufficient to define a dominant combination because stability depends on interactions between the oversampling procedure, the dataset geometry, and the classifier inductive bias. Therefore, the most reliable stability gains are obtained by selecting a more stable classifier family (notably RF), while the choice of oversampling method should be evaluated in a dataset-specific manner rather than treated as a universal stability lever.

4.4 Relationship between stability and predictive performance

Oversampling is typically introduced to improve predictive performance on minority classes, so it is natural to ask whether performance gains are accompanied by more stable explanations, or whether the two objectives can diverge. To study this relationship, we compare, at the dataset level, the average SHAP instability indicator with the corresponding predictive performance measured by the F1-score.

Figure 5 contrasts these quantities across datasets (blue bars for instability, orange bars for F1-score). The figure highlights that the two measures do not co-vary in a systematic manner. Datasets with comparatively high F1-score can still exhibit non-negligible instability, and conversely, low instability does not guarantee superior F1-score. In other words, the relative ordering of datasets by performance differs substantially from the ordering by explanation stability. This observation supports the claim that predictive performance and explanation stability capture different aspects of pipeline behaviour, and that optimising for one does not automatically optimise the other.

From our results, this decoupling is important for auditing and operational monitoring. Oversampling can improve discrimination on minority cases by altering the effective training distribution, but the same intervention can introduce

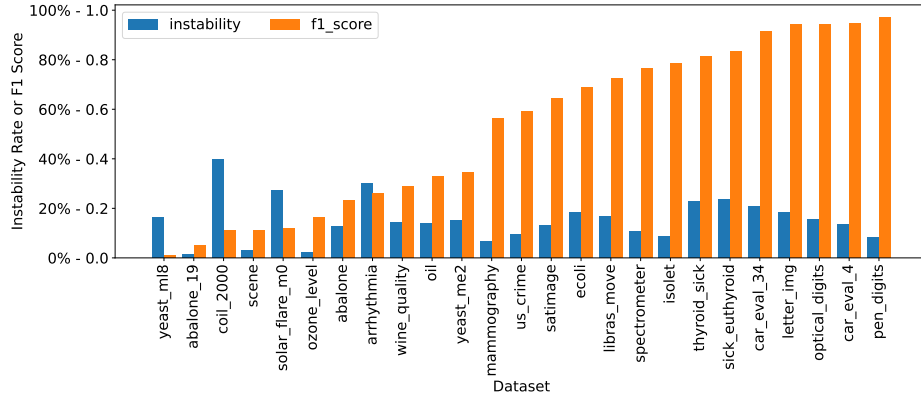


Figure 5: Dataset-level comparison of SHAP instability (blue) versus predictive performance as measured by F1-score (orange). Instability is inversely related to desirability (higher values are worse).

variability in the learned decision structure across random initialisations, which then propagates to explanation variability. Consequently, selecting a pipeline solely based on predictive metrics may lead to configurations that perform well but exhibit explanation drift, which can complicate validation and governance procedures when explanations are used to support model review. Therefore, these results strengthen the conclusion drawn in response to RQ2: stability is better treated as an additional diagnostic objective, reported alongside performance, rather than as a property that can be inferred from predictive metrics alone.

4.5 Relationship between initial dataset imbalance ratio and stability

This subsection analyses how local SHAP explanations change as class balancing is progressively increased through oversampling, complementing the aggregate instability results reported earlier. Considering that our instability indicator captures, for each dataset, classifier, and oversampled scenario, the proportion of test instances whose mean cosine distance between SHAP vectors (computed across different random seeds) increases as the oversampling level moves from 0% to 50%. Therefore, higher values indicate that a larger fraction of instances experiences explanation drift when resampling is intensified.

Figure 6 provides an additional perspective by confirming, at the dataset level, the average SHAP instability (blue bars) with the dataset imbalance ratio (orange bars). The two quantities are shown on the same axis to emphasise their relative behaviour across datasets rather than their absolute scales. This result illustrates two points. First, instability varies substantially across datasets, spanning from near-zero values to markedly higher values, reinforcing that explanation drift under oversampling is not uniformly distributed across problems.

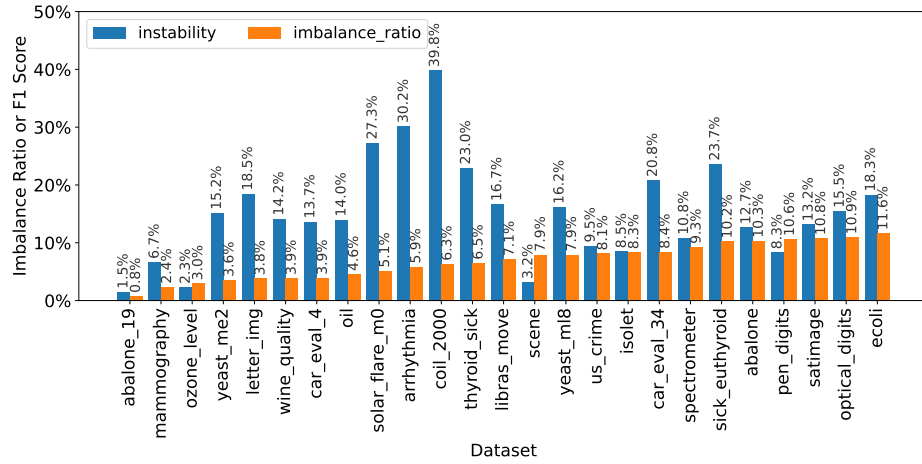


Figure 6: Dataset-level comparison between SHAP instability (blue) and imbalance ratio (orange). Instability corresponds to the proportion of instances whose mean cosine distance between SHAP vectors increases as oversampling intensity grows (higher is worse). The absence of a consistent co-variation indicates that imbalance ratio alone is not a reliable predictor of explanation drift under stronger oversampling.

Second, the ordering of datasets by imbalance ratio does not match the ordering by instability: some strongly imbalanced datasets exhibit relatively modest instability, while several datasets with lower imbalance ratios still present high instability. This evidence supports the conclusion that the effect of oversampling intensity on local SHAP explanations cannot be inferred from class imbalance alone.

4.6 Dataset-driven explanation instability

To address RQ3, we relate the observed SHAP instability under oversampling to intrinsic dataset characteristics expressed as meta-features (Section 3.2). For each dataset, we first compute a dataset-level instability score, and we then extract a comprehensive set of descriptive meta-features. We quantify association using Spearman’s rank correlation, which is suitable for capturing monotonic relationships without assuming linearity.

The correlation analysis shows that instability is primarily associated with the structural properties of the feature space rather than with the class imbalance ratio alone. In particular, the strongest positive correlations are observed for meta-features that capture frequent low-order co-occurrence patterns (e.g., `one-itemset` and `two-itemset` descriptors) and sparsity-related measures. Conversely, instability is negatively correlated with average attribute entropy (`attr_ent.mean`), suggesting that datasets with more diverse attribute distri-

butions yield more stable SHAP explanations under the same resampling perturbations. Figure 7 presents the obtained correlation.

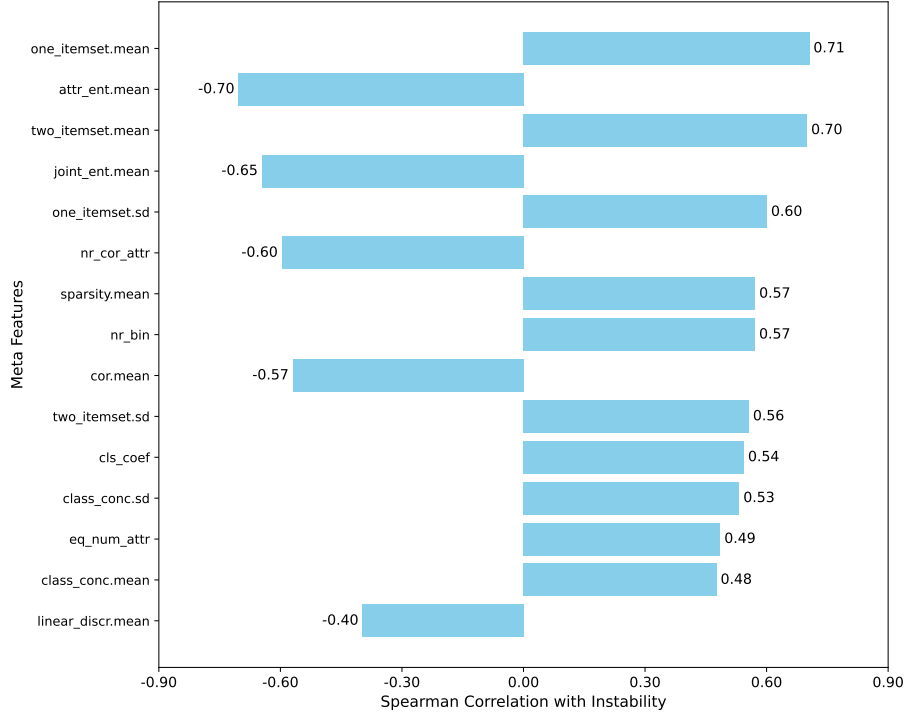


Figure 7: Spearman correlations between dataset meta-features and SHAP instability under oversampling. Large positive coefficients indicate that higher meta-feature values are strongly associated with higher instability, large negative coefficients indicate a strong association with lower instability, and values close to zero suggest little or no monotonic relationship.

Additional negative associations are observed for meta-features capturing inter-attribute correlation (e.g., `cor.mean`, `nr_cor_attr`) and dataset size (e.g., `nr_inst`), suggesting that larger datasets and more redundant global structure tend to dampen run-to-run variability in the learned decision function and, in turn, reduce explanation instability.

Overall, these results answer RQ3, indicating that SHAP instability under oversampling is driven more by dataset structure than by imbalance severity, increasing with sparsity and strong low-order co-occurrence patterns and decreasing with higher attribute and attribute-class entropy, which provides a practical cue to prioritise stability checks for datasets with repetitive, sparse, and structurally concentrated representations.

Table 2: List of meta-features extracted from datasets to compose the surrogate model.

Group	Meta-feature name	Description
general	nr_bin	Compute the number of binary attributes.
complexity	cls_coef	Clustering coefficient.
landmarking	linear_discr	Performance of the Linear Discriminant classifier.
info-theory	attr_ent	Compute Shannon’s entropy for each predictive attribute.
info-theory	class_conc	Compute concentration coefficient between each attribute and class.
info-theory	eq_num_attr	Compute the number of attributes equivalent for a predictive task.
info-theory	joint_ent	Compute the joint entropy between each attribute and class.
statistical	cor	Compute the absolute value of the correlation of distinct dataset column pairs.
statistical	nr_cor_attr	Compute the number of distinct highly correlated pair of attributes.
statistical	sparsity	Compute (possibly normalized) sparsity metric for each attribute.
itemset	one_itemset	Compute the one itemset meta-feature.
itemset	two_itemset	Compute the two itemset meta-feature.

5 Discussion

Our results yield actionable guidance for deploying SHAP in imbalanced learning pipelines:

- *Resampling level is not a monotonic risk factor.* Increasing oversampling does not produce a uniform monotonic change in local explanations across datasets; therefore, stability should be treated as a pipeline property and empirically checked for the target dataset and model family (Section 4.2).
- *Classifier choice is the strongest stability lever.* Stability varies more across classifiers than across oversampling algorithms, with RF showing the lowest aggregate instability, XTR intermediate behaviour, and boosting higher instability, consistent with different sensitivity to training perturbations (Section 4.1).
- *No universally dominant oversampling algorithm.* Differences among oversampling algorithms are smaller and strongly interaction-dependent; although SVMSMOTE is lower on average, the most stable choice depends on dataset geometry and the classifier inductive bias (Section 4.1).
- *Performance does not proxy stability.* The weak alignment between F1-score and instability implies that performance-driven selection can yield pipelines with explanation drift; stability should be reported as a complementary diagnostic when explanations support auditing (Section 4.4).

- *Dataset meta-features provide early warning signals.* Instability is not explained by imbalance ratio alone and is more strongly associated with dataset structure, increasing with sparsity and low-order co-occurrence patterns and decreasing with higher entropy; datasets with these characteristics should be prioritised for stability checks before using SHAP in auditing or decision support (Section 4.6).

6 Limitations and Extensions

This study targets explanation stability under oversampling in imbalanced tabular classification, but the scope imposes limitations. The experiments are restricted to tree-based classifiers and a set of benchmark datasets, so the conclusions may not generalise to other model families or to non-tabular modalities.

The stability analysis also depends on operational choices. We quantify stability using cosine distance between attribution vectors across repeated trainings and by tracking its evolution with increasing resampling intensity. This measure is appropriate for capturing changes in attribution direction, but it is less sensitive to pure magnitude shifts and can be unreliable when attribution norms are very small.

Practical constraints led us to compute explanations on a fixed subset of instances and to estimate variability from a limited number of random seeds; while this enables large-scale coverage, it does not capture additional sources of variation such as alternative train/test splits or cross-validation folds.

Finally, the meta-feature analysis provides associations between dataset properties and instability, but it does not establish causality and is affected by the finite number of datasets relative to the number of candidate descriptors.

These limitations point to several extensions. Future work can broaden the set of learners and imbalance countermeasures, and strengthen the robustness of conclusions by adding complementary stability metrics.

Stability can also be connected to other quality dimensions through tests, ground-truth explanation benchmarks, and lightweight human-centred evaluations.

The observed links between instability and dataset descriptors motivate a meta-learning layer that predicts instability risk from meta-features, providing an a priori warning when oversampling is likely to induce explanation drift.

7 Conclusion

This paper investigates the stability of SHAP explanations in imbalanced classification pipelines that employ oversampling. The findings indicate that common oversampling strategies do not systematically destabilize SHAP explanations. In terms of specific dynamics, we did not observe a uniform monotonic relationship between oversampling intensity and explanation stability; likewise, no

oversampling algorithm emerged as a consistently dominant choice from a stability standpoint. Furthermore, stability can vary substantially across datasets and classifier families.

To clarify when instability is more likely to occur, we complemented the experimental results with a dataset meta-feature analysis. The observed associations are not explained by imbalance ratio alone but by structural properties of the data: instability increases in datasets with stronger low-order co-occurrence patterns and higher sparsity, and decreases with higher attribute and attribute and class entropy, with additional stabilising associations linked to larger sample size and stronger attribute redundancy.

Future work can extend these results by considering additional model families and imbalance countermeasures, validating conclusions under multiple data splits and alternative stability metrics, and developing meta-learning tools that predict instability risk from dataset descriptors to support explanation-aware pipeline selection.

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